

Priya Goyal

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EMPLOYMENT

DatologyAI, Founding Member of Technical Staff

Jan 2024 - Present

Founding Researcher

Leading research on improving data quality (via pruning) to yield more efficient and higher quality language models.

Google Deepmind

Aug 2022 – Jan 2024, New York, NY

Staff Research Engineer

Aug 2022 – Jan 2024

- Leading efforts on building fundamental Multimodal technology aimed at enabling new capabilities with the most capable LLMs with focus on retrieval augmentation, tool use, complex vision tasks, evals, fairness and bias mitigation etc.

Facebook AI Research

Oct 2016 – Aug 2022, New York, NY

Staff Research Engineer

Feb 2021 – Aug 2022

Senior Research Engineer

June 2018 – Feb 2021

Research Engineer

Oct 2016 – June 2018

- Tech Lead of team focused on computer vision and responsible AI research and engineering.
- Extensive research experience in computer vision and deep learning – 16K plus citations, won best paper award at ICCV, led and released libraries (such as VISSL) and organized self-supervised learning challenge at ICCV workshop.
- Large scale vision modeling - built largest (to date) dense computer vision models (10billion parameters) and first author of foundational work – ImageNet in 1hour.

Facebook

Jan 2016 – Sept 2016

Software Engineer

London, UK

- Built deep learning models to improve the knowledge and entities graph of Facebook. Led the work on building neural networks from highly noisy data.

EDUCATION

Indian Institute of Technology (IIT) Kanpur

India, 2010 - 2015

Bachelor and Masters, Mathematics and Scientific Computing

M.Sc. Advisor: Prof. Harish Karnick

Thesis Title: Study of auto-encodings in the hidden units of deep neural networks.

GPA: 9.1/10 (Ranked 2 of 50 in graduating class).

RESEARCH INTERESTS

Socially Responsible / Safe AI: diagnosing and mitigating harms, model privacy and robustness.

Large scale models: pushing the limits in terms of data and model scale.

Multimodal learning: learning with many modalities including text, images.

Data quality / curation: improving data quality for llms and multimodal models.

Computer Vision / Machine Learning: learning with limited supervision (from uncurated data).

Retrieval Augmentation: effective ways to address challenges in LLM, enable new capabilities etc.

PUBLICATIONS

- *Vision Models Are More Robust And Fair When Pretrained On Uncurated Images Without Supervision* - Priya Goyal, Quentin Duval, Isaac Seessel, Mathilde Caron, Ishan Misra, Levent Sagun, Armand Joulin, Piotr Bojanowski, **arXiv 2022**
- *Fairness Indicators for Systematic Assessments of Visual Feature Extractors* - Priya Goyal, Adriana Romero Soriano, Caner Hazirbas, Levent Sagun, Nicolas Usunier, **arXiv, 2022 (under review)**
- *A Self-Supervised Descriptor for Image Copy Detection* - Ed Pizzi, Sreya Dutta Roy, Sugosh Nagavara Ravindra, Priya Goyal, Matthijs Douze, **CVPR 2022**
- *Fully Sharded Data Parallel: faster AI training with fewer GPUs* - Myle Ott, Sam Shleifer, Min Xu, Priya Goyal, Quentin Duval, Vittorio Caggiano, **Facebook Engineering blog, 2021**
- *VISL: A library for state-of-the-art self-supervised learning from images* - Priya Goyal, Quentin Duval, Jeremy Reizenstein, Matthew Leavitt, Min Xu, Benjamin Lefaudeaux, Mannat Singh, Vinicius Reis, Mathilde Caron, Piotr Bojanowski, Armand Joulin, Ishan Misra, **2021**
- *Self-supervised pretraining of visual features in the wild* - Priya Goyal, Mathilde Caron, Benjamin Lefaudeaux, Min Xu, Pengchao Wang, Vivek Pai, Mannat Singh, Vitaliy Liptchinsky, Ishan Misra, Armand Joulin, Piotr Bojanowski, **arXiv 2021**
- *Unsupervised Learning of Visual Features by Contrasting Cluster Assignments* - Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, Armand Joulin, **NeurIPS 2020**
- *Scaling and Benchmarking Self-Supervised Visual Representation Learning* - Priya Goyal, Dhruv Mahajan, Abhinav Gupta*, Ishan Misra*, **ICCV 2019**
- *The next 700 accelerated layers: From mathematical expressions of network computation graphs to accelerated gpu kernels, automatically* - Nicolas Vasilache, Oleksandr Zinenko, Theodoros Theodoridis, Priya Goyal, Zachary Devito, William S Moses, Sven Verdoolaege, Andrew Adams, Albert Cohen, **ACM TACO 2019**
- *Focal Loss for Dense Object Detection* - Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, Piotr Dollár, **ICCV 2017 (best student paper award)**
- *Accurate, Large Minibatch SGD: Training ImageNet in 1 Hour* - Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyrola, Andrew Tulloch, Yangqing Jia, Kaiming He, **arXiv 2017**

Tech Reports

- Peer-to-peer insult detection in online communities - Priya Goyal, Gaganpreet Singh Kalra, 2013

WORKSHOP CHALLENGE

Facebook AI Self-Supervision Learning Challenge Extreme Computer Vision Workshop

ICCV, 2019
Seoul, Korea

Organized the first ever [self-supervised challenge](#) aimed at benchmarking representation learning on a variety of tasks. Wrote the transfer learning code and setup the challenge evaluation/submission server.

INVITED TALKS

- SEER, a self-supervised billion parameter computer vision model - [Facebook AI Innovation Summit, 2021](#)
- Scaling and Benchmarking Self-Supervised learning - Carnegie Mellon University, April 2019.
- Facebook North America Women in AI Leadership Summit, 2019 - 10K attendees.
- ImageNet-1k training in 1 hour - Caffe2 Meetup, Hawaii, CVPR 2017
- ImageNet in 1 hour - Deep Learning at Supercomputer Scale, NeurIPS 2017

MEDIA COVERAGE

- [TechCrunch article on ImageNet in 1-Hour.](#)
- [CNBC article on SEER \(training A.I. to "see"\).](#)
- [NVIDIA Developer on ImageNet on 1-Hour.](#)
- [Geekwire on ImageNet in 1-Hour.](#)
- [NVIDIA Developer on Self-supervised learning beating SOTA Computer vision models.](#)
- [WIRED article on AI Teaching Itself to See With Less Human Help.](#)
- [CNET on training computers to learn like humans do.](#)
- [ImageNet in 1-Hour at NeurIPS 2017 Supercomputing workshop.](#)

AWARDS

- Best Student Paper Award, ICCV 2017.
- Facebook Grace Hopper Scholarship, 2014 and Google Global Scholarship, 2013.
- Academic Excellence Award 2011-12 and 2012-13 – given to top 5% of 850 students.
- Best Poster Award (amongst 60 posters) presented in SURGE, IIT Kanpur 2012.
- SURGE undergraduate research fellowship (2012), Indian Institute of Technology (IIT), Kanpur, India.
- National Talent Search Scholarship, 2008 – given to top 0.01% high school students all over India.

ACADEMIC SERVICE

Conference reviewer: Computer Vision and Pattern Recognition (2018 – present), International Conference on Computer Vision (2018 – present), European Conference on Computer Vision (2018 – present), International Conference on Learned Representations (2018 – present), International Conference on Machine Learning (2018 – present), Neural Information Processing Systems (2018 – present)

INTERNSHIPS

Facebook

Advisor: Kay Rottmann, Juan Miguel Pino, Language Technology Team

May-August 2015

Menlo Park, CA, USA

Built profanity detection classifier to detect profanity in the machine translations produced by Facebook. Built classifier for all the languages and automated the re-training of any classifier.

Goldman Sachs

Advisor: Andrei Bergners, Panna Pavangadkar, Developer Practices Group

May-July 2014

New York, US

Built software for making the firm approved SDLC build cloud dynamic and self-aware. The product analyzes nearly 3000 product builds across the firm on daily basis.

Yahoo!

Advisor: Bennet Manuel, Yahoo! Right Media Exchange

May-July 2013

Bangalore,

India

Built visualization tool for tracking the usage of web bugs in online display ads to detect malicious vendors - helped increase the revenue of the company by curbing ban on minor defaulters.

Carnegie Mellon University Winter School

Advisor: Prof. Carolyn P. Rosé, Department of Computer Science, CMU

Dec 2012-Jan 2013

Develop an interactive tutor moderated chat system for automated analysis and dynamic support for an online collaborative learning process. Proposed and implemented 2-pass architecture model for modelling the students' conversation and building an agent prototype capable of detecting distraction/ confusion among students and provide necessary support to ensure effective learning.

Summer Undergraduate Research Grant for Excellence

Advisor: Prof. Akash Anan, Department of Mathematics and Statistics

May-July 2012

IIT Kanpur,

India

Developed efficient algorithms for projecting the point cloud on the projection surface. Automated the process of 3-D surface generation.